Predicting a deflection from an impact

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate a velocity change and explain the value of lead time.

SUCCESS CRITERIA - I can...

1. State how you simplify the collision.
2. Connect delivered momentum to target velocity change.
3. Explain why real deflection includes ejecta and orbital effects.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “deflection” and “lead time” to explain your answer. Sentence starters are optional.

deflection: a change in the direction or path of motion

lead time: time available before the modeled event

impulse: net force acting over time; change in momentum

01 Warm-up: bring back an earlier idea

Two equal **3 kg** carts approach at **5 m/s** in opposite directions and stick. **Find their common final velocity** and explain.

Word help: final = at the end of the interval; subscript f

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\Delta v \approx p_{projectile}/M_{target}$; shift ≈ Δv × time. Simplified calculation: much heavier target; straight-line shift.

Worked reasoning: In a fictional collision, a **2 kg** projectile at **30 m/s** sticks to a stationary **10 kg** target. Final speed is **60/12=5 m/s**. The system includes both objects.

Your turn: Draw the before-and-after system boundary and momentum arrows.

Sentence starter: In my work, I used … to show … . I labeled it … .

Before **After**

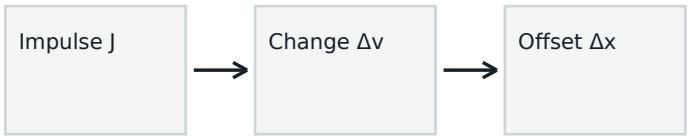
Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

After a brief sideways push, a probe has a constant added sideways velocity. Assume no further sideways force. **Compare its sideways distance after** a short drift and a long drift. **Use the diagram** to connect impulse, velocity change, and later position.

Sentence starter: Both … . They differ in … because … .



Marzano check - circle: ① ② ③

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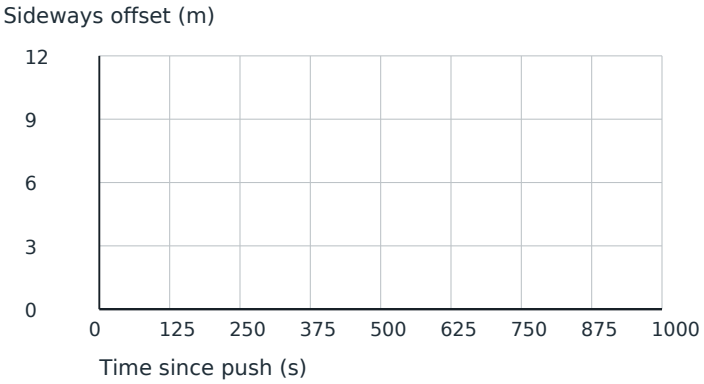
Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A simulated probe receives a brief push that adds **0.01 m/s** sideways to its velocity. It then drifts without further sideways force. **Complete the graph** and **find its sideways offset after 100 s** and **1000 s**.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: ① ② ③

05 Explain your reasoning

Why should the no-ejecta sticking calculation not be reported as the exact measured result of a real mission?

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

An asteroid's period is **6 years**. **Explain why** that number alone cannot establish an Earth encounter. **Name two additional observations needed**.
Organize the timing information before drawing a conclusion.

Word help: encounter = objects reaching nearby positions at the same time

Sentence starter: I think … . I can point to …, which shows …, so … .

Object	Arrival time / window	Comparison
A	____	____
B	____	____

Marzano check - circle: ① ② ③

07 Exit check

At the same small Δv, compare shift with twice as much lead time. **State one thing this prediction** leaves out. Organize the known quantities on the before-and-after diagram.

Sentence starter: Both … . They differ in … because … .

Before	**After**

Label objects and signed velocities.

Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

Return to the worked example in problem **2**. Keep projectile momentum **60 kg·m/s** but double the stationary target mass from **10** to **20 kg**. **Calculate the new final speed** and **explain the change**. Organize the known quantities on the before-and-after diagram.

Sentence starter: I think … . I can point to …, which shows …, so … .

Before	**After**

Label objects and signed velocities.

Marzano check - circle: ① ② ③